



ADAM
SHAW

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PROFESSIONAL SUMMARY

Third-year undergraduate at USC studying Applied Mathematics with a Computer Science minor, concurrently pursuing a master's degree in Statistics. Proficient in data science and visualization tools such as Scikit-learn, Pandas, NumPy, and Seaborn, with experience building deep neural networks with PyTorch. Currently conducting research in topological deep learning, exploring the intersection of algebraic topology and machine learning architectures. Passionate about ML research and hoping to attend a PhD program to dive deeper into the field.

SKILLS

- Python
- PyTorch
- Scikit-learn
- Pandas
- Seaborn
- C++

EDUCATION

B.S., Applied Mathematics; Computer Science Minor
M.S., Statistics
University of Southern California, Los Angeles, CA
Cumulative GPA: 3.7/4.0

Expected to Graduate December 2027

EXPERIENCE

Student Researcher

August 2025 - Present

University of Southern California | Los Angeles, CA

- Conducting directed research with Professor Alvin Jin in the field of topological deep learning
- Investigating the way the topology of data (i.e., Betti numbers and other invariants) changes as it passes through successive layers of a deep neural network
- Working specifically with [predictive coding networks](#) (an alternative to backpropagation), attempting to extend the results of [Naitzat et al.](#) to this novel learning framework

Associate Software Developer Intern

May 2025 - August 2025

Google | New York, NY

- Joined the Workspace AI Platform team, which integrates Gemini into products such as Google Docs, Slides, and Gmail
- Streamlined the creation of custom Gems (personalized versions of Gemini) for internal use and sharing with teams
- Developed a flexible access control system, allowing creators to specify user access to these Gems
- Navigated the largest monorepo in the world, using C++ and an internal server framework to deliver the correct Gems to users based on their client application and access permissions

Student Researcher

December 2024 - May 2025

University of Southern California | Los Angeles, CA

- Worked with Professor James Alcala on the identification and classification of bias in text containing hyperbolic and sarcastic language
- Developed a vector score consisting of the principal components of bias, interpreted as left or right-leaning, for or against some issue, etc.